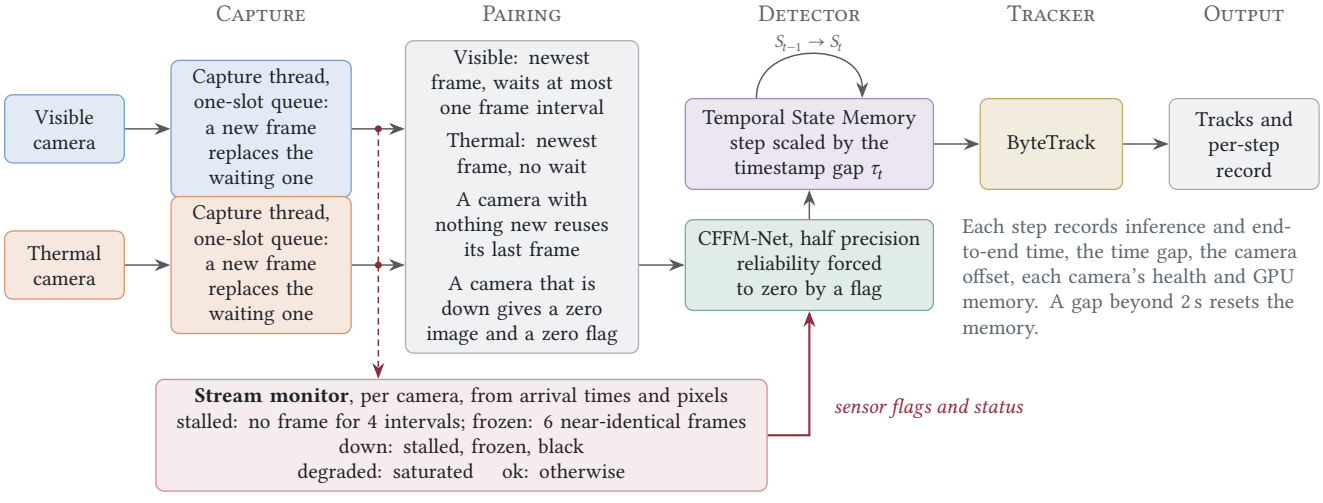
Capture
Pairing
Detector
Tracker
Output
Visible camera
Thermal camera
Capture thread, one-slot queue: a new frame replaces the waiting one
Capture thread, one-slot queue: a new frame replaces the waiting one
Visible: newest frame, waits at most one frame interval
Thermal: newest frame, no wait
A camera with nothing new reuses its last frame
A camera that is down gives a zero image and a zero flag
$S_{t-1} \to S_t$
Temporal State Memory step scaled by the timestamp gap τ_t
CFFM-Net, half precision reliability forced to zero by a flag
ByteTrack
Tracks and per-step record
Each step records inference and end-to-end time, the time gap, the camera offset, each camera's health and GPU memory. A gap beyond 2 s resets the memory.
Stream monitor, per camera, from arrival times and pixels
stalled: no frame for 4 intervals; frozen: 6 near-identical frames
down: stalled, frozen, black
degraded: saturated ok: otherwise
sensor flags and status
Soak test: 30 and 20 frames per second, frames dropped at capture; delay drift, GPU memory growth, throughput
Link-fault run: cable pull, frozen visible, stalled thermal; monitor on (flags fed to the model) against off